

- RAG Learning

2.5 Equations & Inequalities

AQA GCSE Mathematics 8300 — Algebra — spec refs A17–A22 — Papers 1 (non-calculator), 2 and 3 (calculator)

2.5.1 Solving Linear Equations (A17)

An **equation** states that two expressions are equal and is true only for particular values of the unknown. To **solve** it, find the value(s) that make it true. A **linear** equation has no powers of x higher than 1.

Golden rule: whatever you do to one side, do exactly the same to the other side (use inverse operations).

Two-step equations and brackets

Worked Example: Solve $4x + 9 = 37$.

Subtract 9: $4x = 28$ Divide by 4: $x = 7$

Worked Example: Solve $3(x - 2) = 18$.

Either expand: $3x - 6 = 18 \Rightarrow 3x = 24 \Rightarrow x = 8$

or divide first: $x - 2 = 6 \Rightarrow x = 8$

Unknown on both sides

Collect the x terms on the side with the *larger* number of x s (this avoids negatives), and the numbers on the other side.

Worked Example: Solve $7x - 4 = 3x + 16$.

Subtract $3x$: $4x - 4 = 16$

Add 4: $4x = 20$

$x = 5$ Check: $7(5) - 4 = 31$ and $3(5) + 16 = 31$ ✓

Equations with fractions

Multiply every term by the lowest common denominator to clear the fractions. Put brackets round numerators.

Worked Example: Solve $\frac{2x - 1}{3} = \frac{x + 4}{2}$.

Multiply both sides by 6: $2(2x - 1) = 3(x + 4)$

$4x - 2 = 3x + 12$

$x = 14$

Worked Example: Solve $\frac{x + 1}{2} + \frac{x - 2}{3} = 4$.

Multiply every term by 6: $3(x + 1) + 2(x - 2) = 24$

$3x + 3 + 2x - 4 = 24 \Rightarrow 5x - 1 = 24 \Rightarrow 5x = 25 \Rightarrow x = 5$

Approximate solutions from a graph

The solution of an equation such as $2x + 1 = 6$ is the x -coordinate where the line $y = 2x + 1$ meets the line $y = 6$. Reading from a graph gives an **approximate** solution — state it to a sensible accuracy (usually to the nearest small square).

Exam Tip: Always show the inverse operation on each line. A wrong final answer with clear correct working still earns method marks. Check your answer by substituting back into the original equation.

2.5.2 Forming and Solving Equations (A21)

Many questions are set in context: you must **translate** the words into algebra, **solve**, and then **interpret** the answer in the context (with units).

1. Choose a letter for the unknown (say what it stands for).
2. Write expressions for the quantities in the question.
3. Use a fact to form an equation (angles in a triangle sum to 180° , perimeter, total cost, ages...).
4. Solve, then answer the actual question asked.

Worked Example: The angles of a quadrilateral are x° , $2x^\circ$, $(3x - 20)^\circ$ and $(x + 30)^\circ$. Find the largest angle.

Angles in a quadrilateral sum to 360° : $x + 2x + 3x - 20 + x + 30 = 360$

$$7x + 10 = 360 \Rightarrow 7x = 350 \Rightarrow x = 50$$

Angles: 50° , 100° , 130° , 80° . Largest angle = **130°** .

Worked Example: Ben is three times as old as Amy. In 5 years' time the sum of their ages will be 46. How old is Amy now?

Let Amy be a , so Ben is $3a$. In 5 years: $(a + 5) + (3a + 5) = 46$

$$4a + 10 = 46 \Rightarrow 4a = 36 \Rightarrow a = 9. \text{ Amy is } \mathbf{9} \text{ years old.}$$

Exam Tip: Remember to add the time to *both* people in age problems, and to finish by answering the question — a question asking for "the largest angle" is not answered by " $x = 50$ ".

2.5.3 Quadratic Equations (A18)

A **quadratic equation** can be written as $ax^2 + bx + c = 0$ with $a \neq 0$. It can have **two**, **one** (repeated) or **no** real solutions (roots). The roots are where the graph $y = ax^2 + bx + c$ crosses the x -axis.

Solving by factorising

If two things multiply to give zero, at least one of them must be zero. So rearrange to " $= 0$ ", factorise, then set each bracket equal to zero.

Worked Example: Solve $x^2 - 9x + 20 = 0$.

Two numbers that multiply to $+20$ and add to -9 : -4 and -5 .

$$(x - 4)(x - 5) = 0 \Rightarrow x = 4 \text{ or } x = 5$$

Worked Example: Solve $x^2 + 5x = 0$.

Common factor: $x(x + 5) = 0 \Rightarrow x = 0$ or $x = -5$

(Do **not** divide by x — you would lose the solution $x = 0$.)

Equations needing rearrangement Higher

Worked Example: Solve $x^2 = 5x + 14$.

Rearrange to $= 0$: $x^2 - 5x - 14 = 0$

$$(x - 7)(x + 2) = 0 \Rightarrow x = 7 \text{ or } x = -2$$

Worked Example: Solve $2x^2 + 5x - 3 = 0$.

$ac = -6$: find numbers multiplying to -6 and adding to 5 : 6 and -1 .

$$2x^2 + 6x - x - 3 = 2x(x + 3) - 1(x + 3) = (2x - 1)(x + 3) = 0$$

$$x = \frac{1}{2} \text{ or } x = -3$$

Completing the square

Higher

$$x^2 + bx + c = \left(x + \frac{b}{2}\right)^2 - \left(\frac{b}{2}\right)^2 + c$$

Halve the coefficient of x to get the number in the bracket, then subtract its square.

Worked Example: Solve $x^2 - 6x + 4 = 0$ by completing the square, giving answers in surd form.

$$(x - 3)^2 - 9 + 4 = 0 \Rightarrow (x - 3)^2 = 5$$

$$x - 3 = \pm\sqrt{5} \Rightarrow x = 3 \pm \sqrt{5}$$

When $a \neq 1$, take a out as a factor of the x^2 and x terms first: $2x^2 - 12x + 5 = 2(x^2 - 6x) + 5 = 2[(x - 3)^2 - 9] + 5 = 2(x - 3)^2 - 13$.

The quadratic formula

Higher

For $ax^2 + bx + c = 0$:

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

The **discriminant** $b^2 - 4ac$: positive $\Rightarrow$ two roots; zero $\Rightarrow$ one repeated root; negative $\Rightarrow$ no real roots.

Worked Example: Solve $2x^2 - 5x - 4 = 0$, giving answers to 2 decimal places.

$$a = 2, b = -5, c = -4$$

$$x = \frac{5 \pm \sqrt{(-5)^2 - 4(2)(-4)}}{2(2)} = \frac{5 \pm \sqrt{25 + 32}}{4} = \frac{5 \pm \sqrt{57}}{4}$$

$$x = \mathbf{3.14} \text{ or } x = \mathbf{-0.64} \text{ (2 d.p.)}$$

Exam Tip: If a question says "give your answers to 2 decimal places" or "in surd form", the quadratic will not factorise — go straight to the formula or completing the square. Put negative values of b and c in brackets when substituting; $-b$ when $b = -5$ is $+5$.

Approximate solutions from a graph

The solutions of $ax^2 + bx + c = 0$ are the x -coordinates where $y = ax^2 + bx + c$ crosses the x -axis. To solve $x^2 - 2x - 4 = 0$ graphically, read off where the curve cuts $y = 0$: approximately $x \approx -1.2$ and $x \approx 3.2$.

2.5.4 Simultaneous Equations (A19)

Simultaneous equations are two equations in two unknowns that are both true at the same time. The solution is a pair of values (an x and a y) — graphically, the point where the two graphs **intersect**.

Elimination method

Make the coefficients of one variable the same size, then add (if the signs are different) or subtract (if the signs are the same) to eliminate it.

Worked Example: Solve $2x + 3y = 13$ and $5x - 2y = 4$.

Multiply the first by 2: $4x + 6y = 26$

Multiply the second by 3: $15x - 6y = 12$

Add (signs of y are different): $19x = 38 \Rightarrow x = \mathbf{2}$

Substitute: $2(2) + 3y = 13 \Rightarrow 3y = 9 \Rightarrow y = \mathbf{3}$

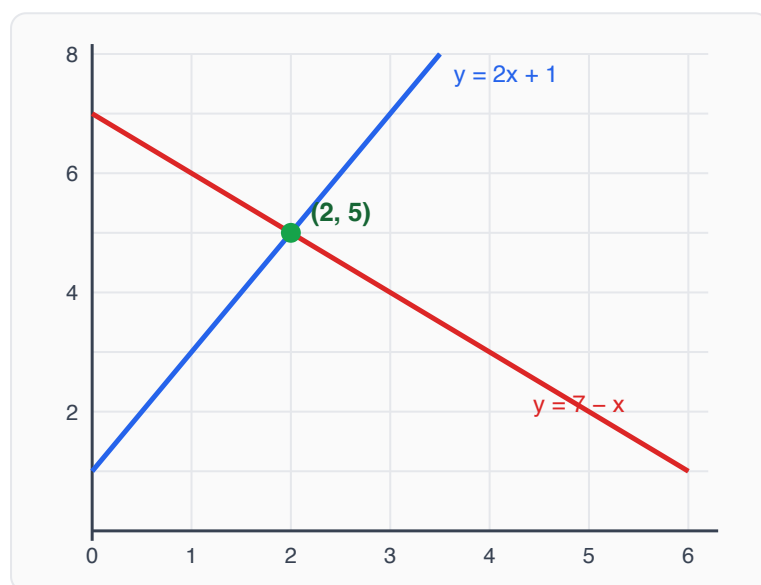
Check in the second equation: $5(2) - 2(3) = 4 \checkmark$

Substitution method

Worked Example: Solve $y = 3x - 1$ and $2x + y = 9$.

Substitute: $2x + (3x - 1) = 9 \Rightarrow 5x = 10 \Rightarrow x = 2, y = 3(2) - 1 = 5$

Graphical solution



The lines $y = 2x + 1$ and $y = 7 - x$ cross at $(2, 5)$, so the solution of the simultaneous equations is $x = 2, y = 5$. Graphical solutions are approximate unless the intersection is on grid points.

Linear and quadratic simultaneous equations Higher

Rearrange the linear equation to make x or y the subject, **substitute** it into the quadratic, solve the resulting quadratic, then substitute each answer back into the *linear* equation to find the partner values. There are usually **two pairs** of solutions (two intersection points of a line and a curve).

Worked Example: Solve $y = 2x - 1$ and $y = x^2 - 4x + 4$.

$$x^2 - 4x + 4 = 2x - 1 \Rightarrow x^2 - 6x + 5 = 0 \Rightarrow (x - 1)(x - 5) = 0$$

$$x = 1 \Rightarrow y = 1; \quad x = 5 \Rightarrow y = 9$$

Solutions: **(1, 1)** and **(5, 9)**

Worked Example: Solve $x^2 + y^2 = 25$ and $y = 2x - 5$.

$$x^2 + (2x - 5)^2 = 25 \Rightarrow x^2 + 4x^2 - 20x + 25 = 25 \Rightarrow 5x^2 - 20x = 0$$

$$5x(x - 4) = 0 \Rightarrow x = 0 \text{ or } x = 4$$

$$x = 0 \Rightarrow y = -5; \quad x = 4 \Rightarrow y = 3. \text{ Solutions: } \mathbf{(0, -5)} \text{ and } \mathbf{(4, 3)}$$

Exam Tip: Pair your answers clearly, e.g. " $x = 1, y = 1$ and $x = 5, y = 9$ ". Always substitute back into the linear equation — substituting into a quadratic like $x^2 + y^2 = 25$ gives extra, wrong partner values.

2.5.5 Iteration Higher (A20)

Iteration finds an approximate solution to an equation by repeatedly using a formula of the form $x_{n+1} = g(x_n)$. Start with x_1 , substitute to get x_2 , substitute x_2 to get x_3 , and so on. The values usually get closer and closer to a solution. The suffix n tells you the position: x_{n+1} is the term after x_n .

Showing a root lies in an interval

If $f(x)$ changes sign between $x = a$ and $x = b$ (and the graph is continuous), then $f(x) = 0$ has a root between a and b .

Worked Example: (a) Show that $x^3 - 3x - 5 = 0$ has a solution between 2 and 3.

$x = 2: 8 - 6 - 5 = -3$ $x = 3: 27 - 9 - 5 = 13$. Sign change, so a solution lies between 2 and 3.

(b) Rearrange: $x^3 = 3x + 5 \Rightarrow x = \sqrt[3]{3x + 5}$

(c) Use $x_{n+1} = \sqrt[3]{3x_n + 5}$ with $x_1 = 2$:

$$x_2 = \sqrt[3]{11} = 2.2239 \dots$$

$$x_3 = \sqrt[3]{3(2.2239 \dots) + 5} = 2.2683 \dots$$

$$x_4 = 2.2769 \dots, \quad x_5 = 2.2786 \dots \rightarrow \text{converging to about } \mathbf{2.28}$$

Exam Tip: Use the ANS key on your calculator: type 2, press =, then type the formula using ANS in place of x_n and press = repeatedly. Write down each value to at least 4 decimal places but carry the full value in the calculator.

2.5.6 Linear Inequalities (A22)

An **inequality** compares two quantities using $<$ (less than), $>$ (greater than), $\leq$ (less than or equal to) or $\geq$ (greater than or equal to). Its solution is usually a **range** of values.

Solving linear inequalities

Solve exactly like an equation, **except**: if you multiply or divide both sides by a **negative** number, **reverse** the inequality sign.

Worked Example: Solve $5 - 2x \leq 13$.

Subtract 5: $-2x \leq 8$

Divide by -2 and reverse the sign: $x \geq -4$

Worked Example: n is an integer with $-6 < 2n \leq 8$. List the possible values of n .

Divide every part by 2: $-3 < n \leq 4$

$n = -2, -1, 0, 1, 2, 3, 4$ (not -3 , because the inequality is strict)

Number lines



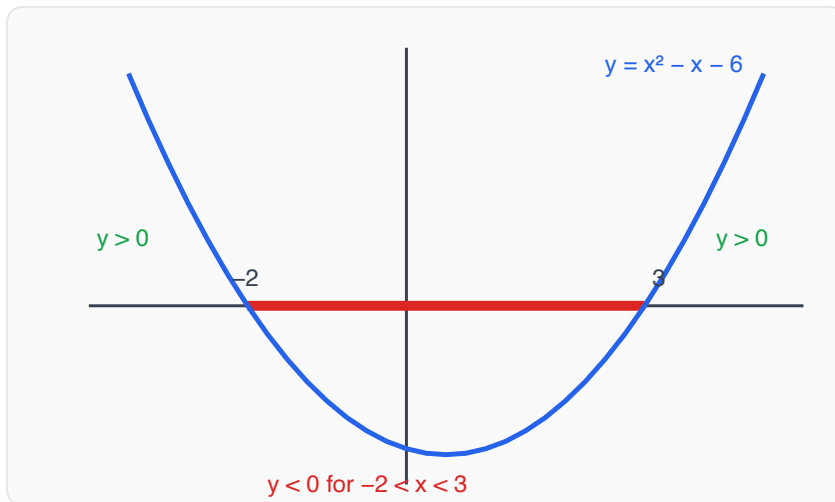
Symbol	Meaning	Number line
$< \text{ or } >$	strict — boundary not included	open (empty) circle
$\leq \text{ or } \geq$	boundary included	closed (filled) circle

Set notation Higher

Solution sets can be written in set notation: $\{x : x > 5\}$ means "the set of all x such that x is greater than 5". Two separate regions are joined with the union symbol: $\{x : x < -2\} \cup \{x : x > 3\}$. A range such as $-1 \leq x < 4$ is written $\{x : -1 \leq x < 4\}$.

2.5.7 Quadratic Inequalities Higher

1. Rearrange so one side is 0 and solve the equation to find the **critical values**.
2. Sketch the u-shaped graph through the critical values.
3. " < 0 " means the part of the curve **below** the x -axis (between the roots); " > 0 " means **above** (outside the roots — two separate inequalities).



Worked Example: Solve $x^2 - x - 6 < 0$.

$(x - 3)(x + 2) = 0$ gives critical values $x = -2$ and $x = 3$.

The curve is below the axis between the roots: $-2 < x < 3$

Worked Example: Solve $2x^2 - 7x - 15 > 0$, giving the answer in set notation.

$(2x + 3)(x - 5) = 0 \Rightarrow x = -\frac{3}{2}$ or $x = 5$

Above the axis outside the roots: $x < -\frac{3}{2}$ or $x > 5$

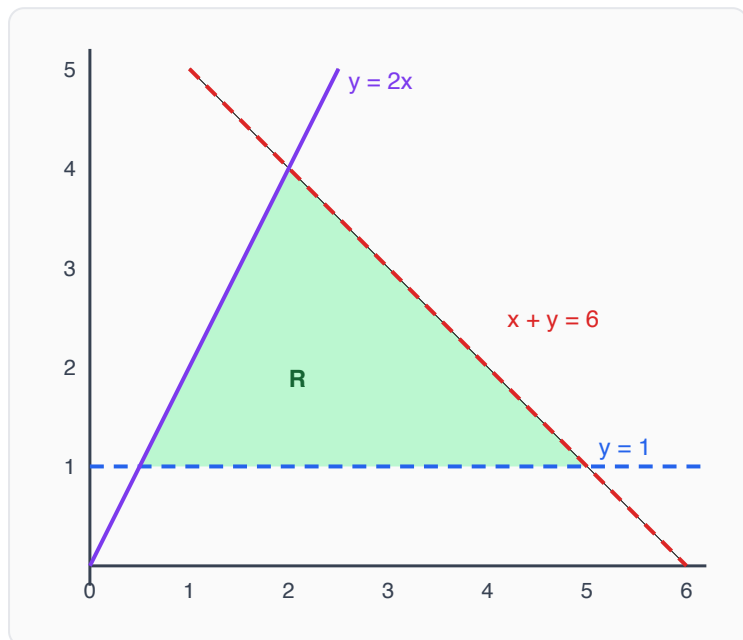
Set notation: $\{x : x < -\frac{3}{2}\} \cup \{x : x > 5\}$

Exam Tip: Never write " $-2 > x > 3$ " for the outside region — no number is both less than -2 and greater than 3 . Write two inequalities joined by "or". Also, $x^2 > 9$ is *not* just $x > 3$: the full answer is $x < -3$ or $x > 3$.

2.5.8 Inequalities in Two Variables (Regions) Higher

An inequality such as $x + y < 6$ describes a **region** of the coordinate grid. Draw the boundary line $x + y = 6$, then decide which side is wanted by testing a point not on the line (the origin is easiest: $0 + 0 < 6$ is true, so the origin's side is the region).

Boundary conventions: dashed line for $<$ or $>$ (points on the line are *not* included); solid line for $\leq$ or $\geq$ (points on the line are included).



Region R satisfies $y > 1$, $x + y < 6$ and $y \leq 2x$. The point $(2, 3)$ lies in R: $3 > 1 \checkmark$, $2 + 3 = 5 < 6 \checkmark$, $3 \leq 4 \checkmark$.

Exam Tip: When asked for integer coordinates in a region, check each candidate point against every inequality, and remember that points on a dashed boundary do not count.

Quick Reference Summary

Type	Method	Tier
Linear equation	Inverse operations; expand brackets; clear fractions by multiplying by the LCD	Foundation
Quadratic (factorisable)	Rearrange to $= 0$, factorise, set each bracket to 0	Foundation ($a = 1$, already $= 0$) / Higher

Quadratic (non-factorisable)	Quadratic formula or completing the square	Higher
Linear simultaneous	Elimination, substitution or graph intersection	Foundation
Linear/quadratic simultaneous	Substitute linear into quadratic; two pairs of answers	Higher
Iteration	Sign change to locate root; $x_{n+1} = g(x_n)$ repeatedly	Higher
Linear inequality	As an equation, reverse the sign when $\times$ or $\div$ by a negative	Foundation
Quadratic inequality	Critical values + sketch; between or outside the roots	Higher
Regions	Boundary lines (dashed/solid), test a point, shade	Higher